Adaptive Regime-Following in Nifty 50: A Meta-Learning Approach

Primary Objective: To design a dynamic, meta-learning-based strategy for the Numin2 dataset to generate discrete daily buy (1), hold (0), or sell (-1) signals.

Core Shift in Paradigm: Moving away from standard continuous price prediction toward Regime-Invariant Statistical Arbitrage.

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Problem Formulation & The Friction Barrier



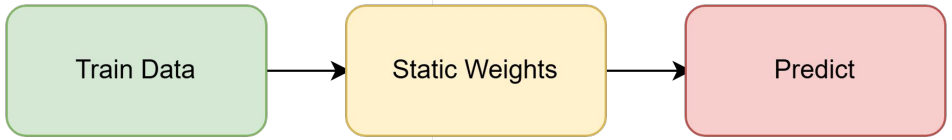


The Non-Stationarity Challenge: Financial markets are highly dynamic, constantly shifting between micro-regimes (e.g., low-volatility bull runs vs. high-volatility crashes). Models trained statically overfit to past regimes and collapse out-of-sample.

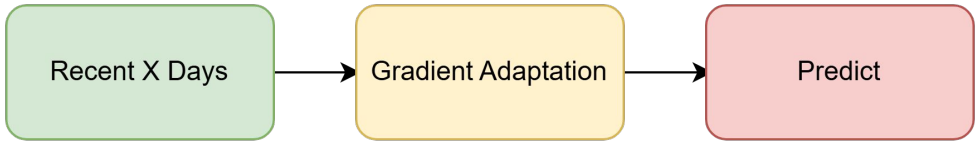
The Transaction Cost Barrier: Real-world trading involves execution friction. Standard predictive models often generate high-turnover signals when uncertain, mathematically destroying portfolio value when transaction fees are applied.

Our Formulation: We frame trading as a sequence of discrete "few-shot" adaptation tasks, requiring the model to rapidly deduce the rules of the *current* market regime.

Baseline Approach vs. Meta-Learning



Baseline



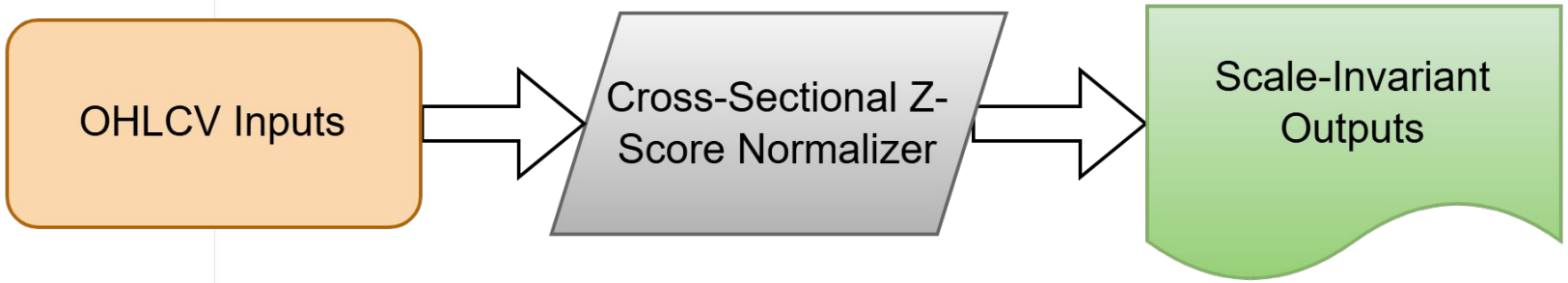
Meta Learning

Baseline Limitations: Standard deep learning baselines utilize "Zero-Shot" inference out-of-sample. They apply static, historically learned weights to new, unseen market conditions, resulting in poor generalization.

The Meta-Learning Advantage: We apply meta-learning concepts to real-world financial market data. Instead of learning rigid rules, the network learns *how to adapt*.

Online Rolling Adaptation: Before predicting the next trading day, our model evaluates the immediate recent past (a rolling window). It takes targeted gradient steps to customize its attention heads specifically to the micro-regime occurring *right now*.

Regime-Invariant Feature Engineering



The Flaw of Raw Data: Standard daily features (Open, High, Low, Close, Volume) force models to anchor to absolute historical price levels, which lose relevance over a 20-year horizon.

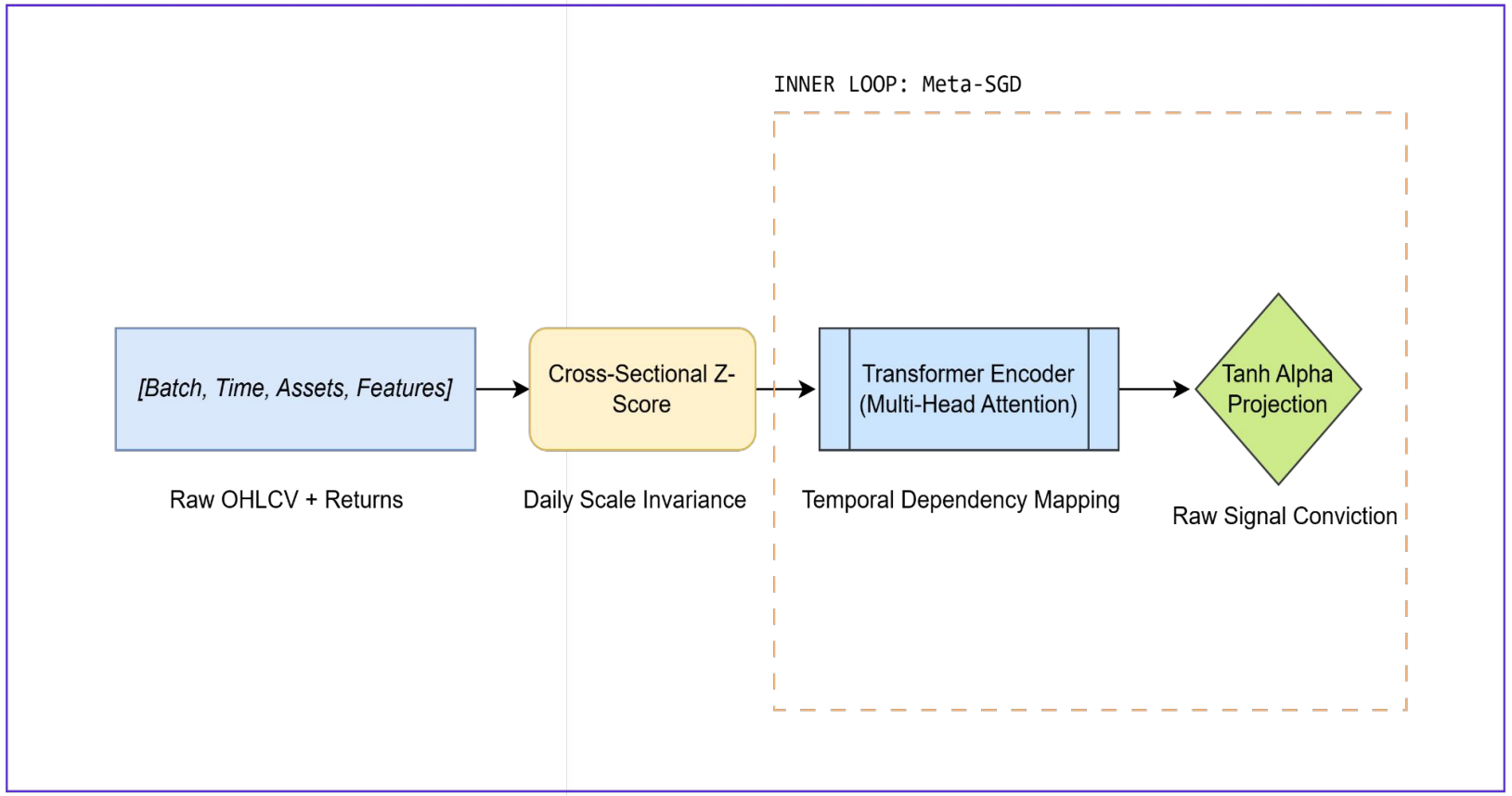
Engineered Alpha Features: We derive complex momentum and volatility indicators (e.g., Garman-Klass Volatility, 60-Day Macro Momentum).

Strict Cross-Sectional Normalization: Every feature is Z-scored daily across the 50-stock universe.

The Result: The network is completely blind to absolute market crashes or rallies. It exclusively evaluates how a stock behaves *relative* to its peers on any given day, isolating pure alpha.

Meta-SGD & SAM Architecture

OUTER LOOP: SAM (Sharpness-Aware Minimization)



- **Backbone:** A Transformer Encoder designed to capture complex cross-asset dependencies over time.
- **Inner Loop (Meta-SGD):** We utilize learnable inner learning rates. The network autonomously discovers *which* specific layers must shift aggressively to adapt to a new regime, and which should remain strictly stable.
- **Outer Loop (SAM):** Sharpness-Aware Minimization optimizes global weights into a "flat minimum," ensuring the base state is highly resilient to noise before fast-adaptation occurs.
- **Objective Alignment:** Instead of MSE, we minimize Pearson Correlation Loss to optimize strictly for the Information Coefficient (cross-sectional ranking).

Work Done & Next Steps

Current Status : The Meta-Learning pipeline and regime-invariant features are compiled and functional.

- An allocator is successfully executing within the Numin2 backtester, actively neutralizing transaction cost bleed.

  Best PnL (52.05)

  Best Sharpe (1.89)

Next Steps for Complete Submission :

- **Optimization of Fast-Adaptation Latency:** Fine-tuning the batch-size for the inner-loop adaptation to determine the optimal historical window (e.g., 20 days vs 60 days) for identifying micro-regime shifts.
- **Expansion of Cross-Sectional Alpha Features:** Integrating additional normalized oscillators (e.g., Cross-Sectional RSI and Mean-Reversion spreads) to provide the Transformer with richer relative-strength context.
- **Allocator Convergence Tuning:** Dynamically adjusting the "Inertia Matrix" coefficients and the Top-K concentration ratio to find the mathematical ceiling of the Sharpe Ratio in high-friction environments.